

# Progressive-Change BACIPS: a flexible approach for environmental impact assessment

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## Summary

1. The Before-After Control-Impact Paired Series (BACIPS) design distinguishes natural spatial and temporal variability from variation induced by an environmental impact (or intervention) of interest. BACIPS is a powerful tool to derive inferences about interventions when classic experimental approaches (e.g. which rely on spatial replicates and random assignment of treatments) are not feasible or desirable. Previously applied BACIPS designs generally assume that effects are sudden, constant and long-lived: that is, that systems exhibit 'step-changes' in response to interventions. However, complex ecological interactions or gradual interventions may create delayed and/or progressive responses, potentially impeding the reliability of classic (step-change) analyses.

2. We propose a novel approach, the Progressive-Change BACIPS, which generalizes and expands the scope of BACIPS analyses. We evaluate the relative performance of this approach using both simulated and real data that exhibit step-change, linear, asymptotic and sigmoid responses following an intervention. We quantify the statistical power and accuracy of the Progressive-Change BACIPS under varying initial population densities, intensity of spatial sampling, effect sizes and number of sampling dates *After* the intervention.

3. We show that Progressive-Change BACIPS identified the correct model among the set of candidate models under most conditions and led to accurate estimates of the parameters that were used to generate the simulated data. When data were sparse, and the dynamics complex, simpler (more parsimonious) models were favoured over the more complex models that actually generated the simulated data. Application of the Progressive-Change BACIPS to existing data sets from the literature led to strong support for specific models (over alternatives) and led to more specific inferences than possible under the classic BACIPS approach.

4. The Progressive-Change BACIPS proposed here is more flexible than the original BACIPS formulation because the data are used to inform the form of the final model, rather than having the form of the model imposed on the data. This leads to better estimates of the effects of environmental impacts and the time-scales over which they operate. As a result, the Progressive-Change BACIPS should be applicable to a wide range of studies and should help improve investigation of time-dependent effects. R code to perform Progressive-Change BACIPS analysis is provided.

**Key-words:** assessment design, Before-After Control-Impact Paired Series, data analysis, effect size, impact assessment, multi-model inference, resources management, simulations, statistical models

## Introduction

Human activities affect ecosystems, and ecologists are often called upon to quantify the impacts of those activities, whether the effects of the interventions are expected to be negative (e.g. an oil spill) or positive (e.g. the establishment of a protected area). In most cases, the interventions are not suited to classic experimental study, owing to the lack of true replicates and the

non-random assignment of sites to 'treatments' (Underwood 1992; Stewart-Oaten 1996; Stewart-Oaten & Bence 2001; Osenberg *et al.* 2006).

The challenge is to compare the state of the system in the presence of the intervention with the state of the system that would have existed if the intervention never occurred. This requires an approach that can reliably estimate the effect of the intervention amid the diverse sources of spatial and temporal variability in the studied system, and this requires data collected before the intervention has occurred (e.g. Stewart-

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Oaten, Murdoch & Parker 1986; Osenberg *et al.* 2006, 2011). A variety of before-data-based approaches have been proposed; each with their own set of strengths and weaknesses (Box & Tiao 1975; Underwood 1990; Stewart-Oaten & Bence 2001). Perhaps the best-known and most powerful approach is the Before-After Control-Impact Paired Series design (BACIPS: Stewart-Oaten, Murdoch & Parker 1986; Osenberg *et al.* 2006). BACIPS has been applied to a wide range of situations, including effects of nuclear power plants on kelp (Bence, Stewart-Oaten & Schroeter 1996), invasive species on invertebrates (Kadye & Booth 2012), river diversion on fish (de Mutsert & Cowan 2012), logging on vertebrates (Crome, Thomas & Moore 1996), or restoration programs on vegetation (Bousquin & Colee 2014) and invertebrates (Suren, Lambert & Sorrell 2011).

In BACIPS designs, Impact and Control sites are sampled simultaneously (or nearly so) multiple times Before and After an intervention. For the  $i$ th sampling survey conducted in period  $P$  (Before or After), the difference in the sampled response variable (e.g. density), or its transformation, is determined:

$$\Delta_{P,i} = N_{\text{Impact},P,i} - N_{\text{Control},P,i} \quad \text{eqn 1}$$

The average difference in the Before period,  $\bar{\Delta}_{\text{Before}}$ , estimates the spatial variation between the sites and comprises a null hypothesis for the differences that would exist in the *After* period, *in the absence of an intervention*. The differencing accounts for spatial variation among sites as well as shared factors whose effects vary through time (i.e. it accounts for spatial and temporal variance in the response variable). Spatiotemporal variation (i.e. unique temporal fluctuations at each site) adds noise to the approach, but this is addressed, in part, by repeated sampling through time (to characterize the magnitude of that variation and its correlation structure; Stewart-Oaten & Bence 2001). The temporal variance of  $\bar{\Delta}_{\text{Before}}$  is driven by the spatiotemporal variation in the response variable, in addition to sampling error, and limits the power of the BACIPS analysis (Osenberg *et al.* 1994, 2006).

In its classic application, a BACIPS analysis compares the Before and After differences using a two-sample test (e.g.  $t$ -test or ANOVA: Stewart-Oaten, Murdoch & Parker 1986; Stewart-Oaten & Bence 2001). As a result, the BACIPS approach asserts that the intervention causes a step-change in the response variable coincident with the start of the intervention. The magnitude of the induced change (estimated as  $\bar{\Delta}_{\text{After}} - \bar{\Delta}_{\text{Before}}$ ) is assumed constant and provides a measure of the effect of the intervention.

However, many interventions may not cause immediate, constant changes to a system. Indeed, complex ecological interactions may take time to propagate through the system, leading to time-dependent effects. Hence, the classic comparison of Before and After differences may not be suitable for most applications. In addition, understanding and quantifying the shape of the response to an intervention is as important as detecting the presence of an intervention. Thus, a more flexible approach is required.

Here, we develop a new statistical approach to the BACIPS design, which we call the Progressive-Change

BACIPS. The Progressive-Change BACIPS fits competing models (including a step-change) to the temporal response of a system following an intervention; it therefore provides a more complete description of the time-dependent changes in the systems arising from an intervention. We evaluate the utility of this approach by simulating data with known dynamical properties and then assessing how well the approach performs when applied to these data. To evaluate the robustness of the approach under different contexts, we also vary the conditions used to generate the data sets. Specifically, we vary the initial state of the ecosystem (i.e. initial density or some other response variable), the type of induced response (i.e. step-change, linear, asymptotic or sigmoid), the magnitude of the change (i.e. the effect size) and the sampling effort used to collect data (i.e. the number of replicate samples within a site and number of sampling dates after an intervention). We also applied the Progressive-Change BACIPS to three real-world data sets originally collected to assess impacts in marine and terrestrial ecosystems.

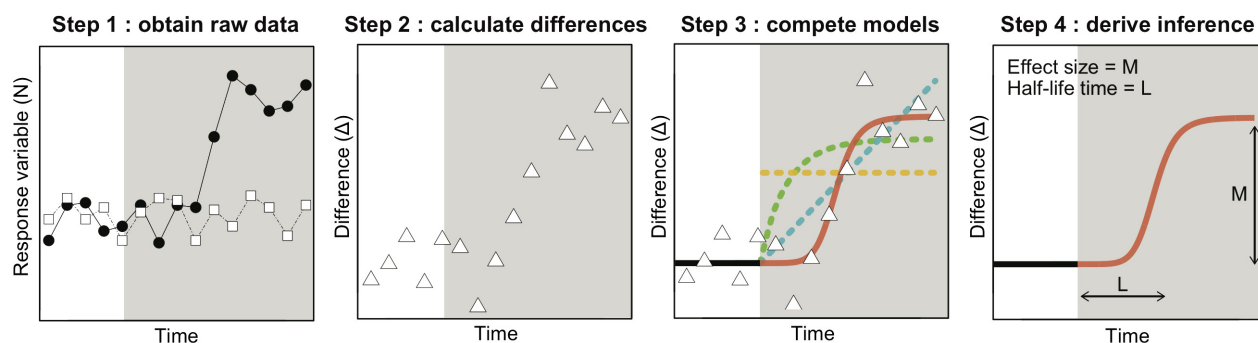
Our ultimate goal is to promote more successful detections and quantifications of human interventions on ecosystems by better matching the inferential tools to the underlying dynamics of the study systems.

## Materials and methods

### PROGRESSIVE-CHANGE BACIPS

We present Progressive-Change BACIPS as a four-step process through which a change in the system is described using multi-model inference (Fig. 1). First, we obtain the raw data by sampling the Impact and Control sites on several dates Before and After the intervention (Fig. 1, Step 1). Secondly, we derive the time series of differences ( $\Delta$ ) between the Impact and Control sites (Fig. 1, Step 2). Thirdly, we compare competing models to the full time series of differences, and the best model is then selected using model selection criteria, such as AIC (Fig. 1, Step 3). Finally, we derive inferences based upon the selected model: that is, the resulting parameter(s) derived from the best model provide(s) an estimate of the ecological effect of the intervention (Fig. 1, Step 4). R code to perform Progressive-Change BACIPS analysis is provided in Appendix S1 (Supporting Information).

Progressive-Change BACIPS builds on the classic BACIPS approach and the extensive research that underlies that tool (Stewart-Oaten, Murdoch & Parker 1986; Osenberg *et al.* 1994; Osenberg *et al.* 2006; Stewart-Oaten & Bence 2001). In all BACIPS studies, the Before time series should be used to evaluate assumptions of the BACIPS model (e.g. evaluate stationarity of Before differences, additivity of time and site effects, autocorrelation; Stewart-Oaten, Bence & Osenberg 1992; Stewart-Oaten 1996). These assumptions are generic to all BACIPS analyses (i.e. they are not unique to the Progressive-Change BACIPS analysis), and they focus on spatiotemporal patterns, especially in the *Before* data. Here, instead, we focus on the challenge of discerning and accurately quantifying effects that arise in the *After* period. For elaboration on the use and importance of the Before data, we refer readers to the seminal work of Stewart-Oaten, Murdoch & Parker (1986), Stewart-Oaten, Bence & Osenberg (1992), Stewart-Oaten (1996) and Stewart-Oaten & Bence (2001).



**Fig. 1.** Steps of the Progressive-Change BACIPS. Step 1: a selected response variable ( $N$ ), such as population density, is simultaneously measured at Impact (black circles) and Control sites (white squares) several times Before (white background) and After (grey background) the intervention; Step 2: the difference ( $\Delta$ ) between the Impact and the Control sites is calculated for each time (white triangles); Step 3: models are fitted to the differences and compared using model selection tools; Step 4: the best model is used to draw inferences about the effect of the intervention.

#### EVALUATION OF PROGRESSIVE-CHANGE BACIPS USING SIMULATED DATA

Successful application of Progressive-Change BACIPS should approximate the actual underlying dynamics of systems following an intervention (i.e. it should return estimates of dynamics and effects consistent with the process that generated the data), and it should be able to do so under a wide range of conditions. To evaluate the approach, we therefore simulated data for Control and Impact sites, Before and After a chronic intervention to a system. After data were simulated assuming four different dynamical responses of the system following the intervention (Table 1): a step-change in the difference (in which the difference arises instantly and is constant through time: that is the case assumed by the classic BACIPS model), a linear change (in which the difference increases through time at a constant rate), an asymptotic change (in which the rate of change decreases monotonically to zero as time passes) and a sigmoid change (in which the rate of change initially increases with time, but eventually decreases to zero). We chose these four functions because they represent qualitatively distinct dynamics that are broadly representative of a range of possible outcomes, although any plausible models could have been chosen.

In each simulation, Control and Impact data were sampled from a Poisson distribution with a fixed initial population density in the Before period ( $N_{\text{Before}}$ ). On each date, a site was sampled with  $R$  replicates (e.g. plots, transects or core samples). For simplicity, we assumed that the true densities in the Impact and Control sites were equal at all times Before the intervention ( $N_{\text{Before,Impact}} = N_{\text{Before,Control}}$ ); therefore, all of the temporal variation in the Before differences arose from sampling error in our simulations (although this effect could also be interpreted as representing spatiotemporal variation in the difference). After the start of the intervention, we assumed that the true density at the Control site remained unchanged from the Before period, but that the density at the Impact site changed based upon the underlying model (step, linear, asymptotic or sigmoid change), which, combined with sampling error, resulted in a time series of differences during the After period.

Armed with this basic approach, we then systematically varied the conditions associated with each simulated scenario. In addition to changing (i) the underlying model (step, linear, asymptotic or sigmoid), we also varied (ii) the Before density ( $N_{\text{Before,Impact}} = N_{\text{Before,Control}} = 1; 10; 100$ ), (iii) the number of replicates per sampling survey ( $R = 1; 10; 100$ ) and (iv) the number of surveys performed during the After period

**Table 1.** Simulated type of responses following an intervention (using the step-change, linear, asymptotic and sigmoid models) and equations describing the difference between the Control and Impact sites

| Model       | Equation                                                                                          | Ecological interpretation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Step-change | $\Delta_{\text{After},i} = M + \Delta_{\text{Before}} + \varepsilon_i$                            | The intervention causes an immediate change (of magnitude $M$ ) in the difference                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Linear      | $\Delta_{\text{After},i} = r t_i + \Delta_{\text{Before}} + \varepsilon_i$                        | The intervention causes a continuous change in the difference that accrues at a constant rate ( $r$ )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Asymptotic  | $\Delta_{\text{After},i} = \frac{M t_i}{L + t_i} + \Delta_{\text{Before}} + \varepsilon_i$        | The intervention causes a continuous change in the difference but with $\Delta_{\text{After},i}$ changing at a declining rate and approaching an asymptote ( $\Delta_{\text{Before}} + M$ ). Thus, $M$ determines the final effect (its magnitude and direction), and $L$ is the half-saturation constant (i.e. the time at which half of the final effect is achieved). The initial slope at $t = 0$ is $M/L$                                                                                                                                                                                                                                                                                                                                                                                           |
| Sigmoid     | $\Delta_{\text{After},i} = \frac{M(t/L)^K}{1 + (t/L)^K} + \Delta_{\text{Before}} + \varepsilon_i$ | The intervention causes a continuous change in the difference, but $\Delta_{\text{After},i}$ shifts from initially accelerating to decelerating. As $t \rightarrow \infty$ , the rate of change in $\Delta_{\text{After},i}$ approaches zero and $\Delta_{\text{After},i}$ approaches its asymptote (at $\Delta_{\text{Before}} + M$ ). We assume that $K > 0$ and $L > 0$ ; thus, $K$ , $M$ and $L$ each contribute to the shape of change, but the sign of $M$ determines the direction of change. The slope at $t = 0$ is 0 (as is the slope as $t \rightarrow \infty$ ). As in the asymptotic model, $L$ is the half-saturation constant: that is when $t = L$ , half of the total effect has been achieved (i.e. $\Delta_{\text{After}} = \Delta_{\text{Before}} + M/2$ ), and the slope is $MK/4L$ |

In all cases, the difference After the intervention ( $\Delta_{\text{After},i}$ ) is expressed as a function of the difference Before ( $\Delta_{\text{Before}}$ ), time since the start of the intervention ( $t_i$ ), and the parameters describing the effect size (e.g.  $M$ ,  $r$ ,  $K$  and  $L$ ). The step-change model corresponds to the classic BACIPS approach (e.g. Stewart-Oaten, Murdoch & Parker 1986; Stewart-Oaten, Bence & Osenberg 1992), whereas the other three models postulate that effects accumulate more gradually. In all models,  $\varepsilon_i$  represents error around the expected difference on the  $i$ th sampling date in the After period.

( $S_{\text{After}} = 5; 10; 20$ ) (Fig. 2). We also varied the effect of the intervention, with the form of this effect depending on the model we used to generate the data. For step-change, asymptotic and sigmoid models, we defined the true effect ( $E_{\text{True}}$ ) as the difference between the true difference Before ( $\Delta_{\text{Before, True}}$ ) and the eventual asymptote ( $\Delta_{\text{Before, True}} + M$ ); hence  $E_{\text{True}} = M$ , which we varied from 0 to 100 in increments of 1. For the linear model, we varied the rate at which the two sites diverged in density ( $E_{\text{True}} = r$ ), which we varied from 0 to 5 in increments of 0.05. For the asymptotic model,  $L$  (which is the half-saturation constant, Table 1) was fixed at 1, and for the sigmoid model,  $K$  and  $L$  (which collectively affect the steepness of the curve and the rate at which the maximum effect is approached, Table 1) were both set to 5. For each simulation, we fixed the number of sampling dates Before the intervention at  $S_{\text{Before}} = 5$ . This yielded a total of 10 908 conditions, each of which was simulated 1000 times.

For each simulation, we then did two things. First, we fit each of the four models (i.e. step, linear, asymptotic or sigmoid, Table 1) to each simulated data set to evaluate whether (and under which conditions) the approach identified the correct model from among the alternates (Fig. 1, Step 3). To fit each model, all surveys from the Before period were assigned  $t = 0$ , and all surveys from the After period were assigned times representing the time since start of the intervention. Thus, we assumed, as it is done in classic BACIPS analyses, that  $\Delta_{\text{Before}}$  is stationary and its variation represents the variation in  $\Delta$  in the absence of the intervention. We used the AICc weight of each model as a measure of their relative likelihood (Akaike 1973; Hurvich & Tsai 1991). From the 1000 simulations, we computed average AICc weights ( $\omega \pm 95\%$  CI) for each model and for each tested condition; if the correct model had a

higher  $\omega$ , then we concluded that the approach was able to discern the true underlying model from the other (incorrect) candidate models.

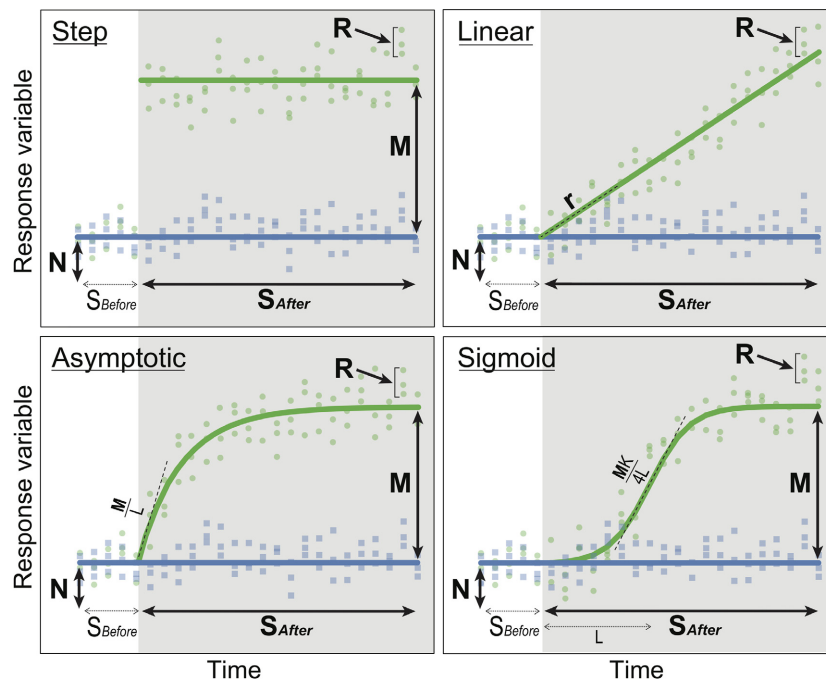
Secondly, given that the approach was generally able to discern the correct model, we next evaluated the ability of the correct model to estimate the true effect size. To achieve this, we applied the Progressive-Change BACIPS to the same simulated data, but only using the correct form of the model (i.e. omitting Step 3; Fig. 1), and used the parameter estimates to derive the effect size (which we refer to as  $E$ , but which corresponds to estimates of  $M$  for the step-change, asymptotic and sigmoid models, and  $r$  for the linear model). The 1000 simulations for each condition yielded an average estimated effect ( $E_{\text{Estim}} \pm 95\%$  CI), which was compared to the true effect ( $E_{\text{True}}$ ).

#### APPLICATION OF PROGRESSIVE-CHANGE BACIPS TO REAL DATA

To provide real-world demonstration of the Progressive-Change BACIPS, we applied it to three data sets from published case studies of environmental impacts.

Barros, Alvarez & Velando (2014) studied the effect of the 2002 *Prestige* oil spill on the reproductive success of the European shag (*Phalacrocorax aristotelis*). The authors annually measured the number of fully grown chicks per nest in oiled (Impact) and unoled (Control) sites over 18 years (1994–2012), which yielded eight unpaired (not used here) and 11 paired annual surveys ( $S_{\text{Before}} = 3$ ,  $S_{\text{After}} = 8$ ).

Torres *et al.* (2011) used a BACIPS design to investigate the consequences of the construction of a four-lane highway on the abundance



**Fig. 2.** Simulated type of responses following an intervention and parameters used to simulate step-change, linear, asymptotic and sigmoid changes at the Impact site. The blue squares (and line) represent the data (and trajectory) for the Control site; the green points (and line) represent the data (and trajectory) for the Impact site. The shaded region denotes the After period. The number of sampling surveys after the intervention ( $S$ ), the response parameter ( $N$ : e.g. density), the number of replicates per sampling survey ( $R$ ), the long-term effect of the intervention ( $M$  for step, asymptotic and sigmoid changes,  $r$  for linear changes) and the time required to achieve half of the long-term effect ( $L$ ) are indicated.  $K$  is a scaling parameter in the sigmoid model that affects the steepness of the curve. For the linear model, the slope of the green line is  $r$  when  $t > 0$ , for the asymptotic model, the slope of the green line is  $M/L$  at  $t = 0$ , and for the sigmoid model, the slope of the green line is  $M/4L$  at  $t = L$ . When performing simulations, we varied parameters indicated in bold ( $r$ ,  $R$ ,  $N$ ,  $S_{\text{After}}$  and  $M$ ) and fixed the others ( $S_{\text{Before}}$ ,  $K$  and  $L$ ).



of the great bustard (*Otis tarda*). Great bustard abundance was compared in an Impact site and two Control sites before, during and after highway construction. Here, we treat the first four samples as Before data ( $S_{\text{Before}} = 4$ ) and the final dates as After data ( $S_{\text{After}} = 8$ ), and use data from only one of the control sites (CZ1).

Williams *et al.* (2009) analysed the effect of Fisheries Replenishment Areas (FRAs, a type of marine protected area) on the abundance of the yellow tang (*Zebrasoma flavescens*) in Hawaii. FRAs were established along the West Hawaii coast in 1999 to support fish recovery from overfishing. Yellow tang density was monitored by underwater visual census in paired areas (FRA sites vs. sites open to fishing) 1 year Before ( $S_{\text{Before}} = 6$ ) and 7 years After implementation of the FRA ( $S_{\text{After}} = 40$ ).

Data were extracted from papers using WEBPLOTDIGITIZER v3.5 (<http://arohatgi.info/WebPlotDigitizer>). All analyses were implemented using the R statistical software version 3.0.1 (R Core Team 2014); MINPACK.LM (Elzhov *et al.* 2013) and NSL2 (Grothendieck 2013) packages were used to perform nonlinear regressions, the AICCMODAVG package (Mazerolle 2016) was used to evaluate second-order Akaike Information Criterion, and the doMC package (Revolution Analytics 2013) was used to increase computational performance.

## Results

### CHOOSING THE BEST MODEL USING SIMULATED DATA

When the underlying process was a step-change, the step-change model always yielded the highest relative likelihood

(AICc weight,  $\omega$ ; Figs 3 and S1 in Appendix S2). Even when the true effect was small, model selection favoured the step-change model.

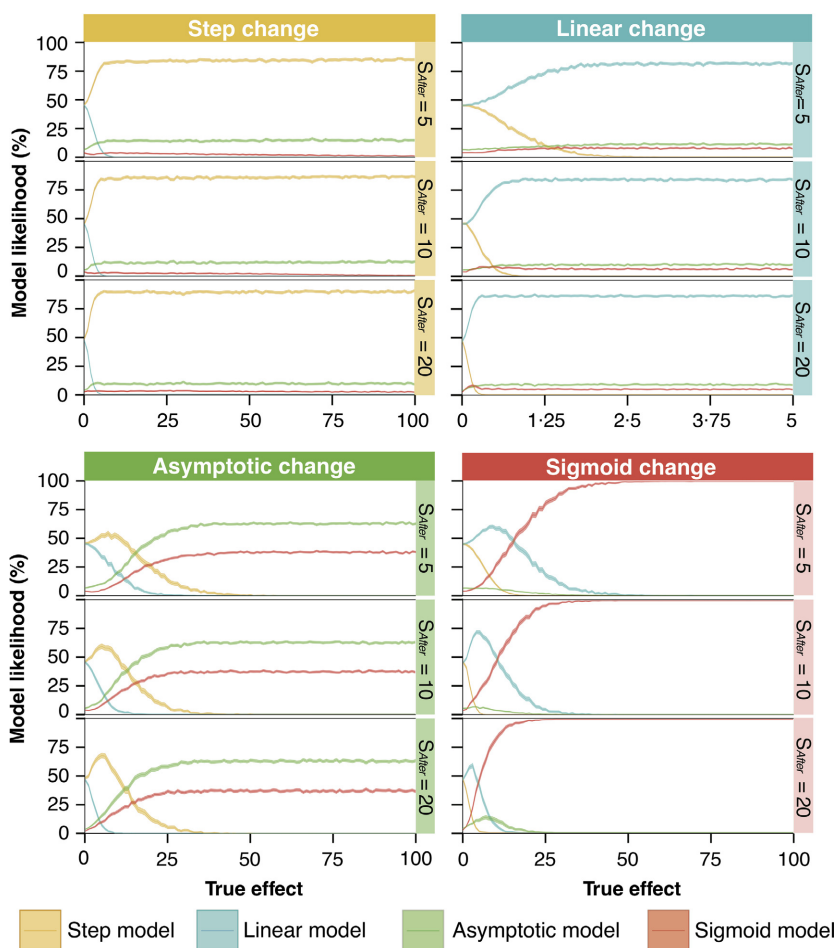
Similarly, the linear model was always the best fitted model when a linear process generated the data (Figs 3 and S2 in Appendix S2).

The asymptotic model was usually the best performing model when the intervention induced an asymptotic response. However, when replication was low (e.g.  $R = 1$ ) or effects were small (e.g.  $E_{\text{True}} < 10$ ), the step and – to a lesser extent – the linear models outperformed the asymptotic model (Figs 3 and S3 in Appendix S2).

The ability to correctly identify the underlying model was the most compromised when the intervention induced a sigmoid response (Figs 3 and S4 in Appendix S2). When effects were small (e.g.  $E_{\text{True}} < 25$ ), replication was low (e.g.  $R = 1$ ), or there were few sampling surveys in the After period (e.g.  $S_{\text{After}} = 5$ ), the linear and – to a lesser extent – the step-change models outperformed the sigmoid model. However, when effects were large, or there was a protracted After period, the sigmoid model was supported over the other models.

### EFFECT SIZE ESTIMATION FROM SIMULATED DATA

The accuracy of the Progressive-Change BACIPS was generally high when the right model was used to estimate effects,



**Fig. 3.** Models competition represented by the relative likelihoods (AICc weights  $\pm$  95% CI) for the four models (step = yellow; linear = blue; asymptotic = green; sigmoid = red) as a function of the magnitude of the true effect generated by the four processes. Each panel gives the average likelihoods based on fitting each of the four models to 1000 simulations for the specific process (step, linear, asymptotic or sigmoid), for different true effects ( $E_{\text{True}}$ : 0–100 for the step, asymptotic and sigmoid cases; 0–5 for the linear case), and number of sampling dates in the After period ( $S_{\text{After}}$ : 5, 10 or 20). All results were obtained for  $N = 10$  and  $R = 10$ ; see Appendix S2 for results obtained by varying  $N$  and  $R$ . For the asymptotic change,  $L = 1$ . For the sigmoid change,  $K = 5$  and  $L = 5$ .

although the accuracy of the estimates of the effects depended upon the underlying model (Fig. 4 and Appendix S3). The step-change model performed well, giving accurate estimates of effects under all tested conditions, even with few replicates ( $R = 1$ ) and few sampling surveys in the *After* period ( $S_{\text{After}} = 5$ ) (Figs 4 and S1 in Appendix S3).

Similarly, the average effect estimated by the linear model (when applied to data generated by a linear process) was very similar to the true effect in all situations, although variability among simulations was slightly higher compared to the results using the step-change model, especially with few sampling dates in the *After* period (Figs 4 and S2 in Appendix S3).

The asymptotic model provided accurate effect estimates when density was low and the number of replicates was high (Figs 4 and S3 in Appendix S3). However, when the *After* time series was short (e.g.  $S_{\text{After}} = 5$  and  $S_{\text{After}} = 10$ ), the number of replicates was small (e.g.  $R = 1$  and  $R = 10$ ), initial density was high (e.g.  $N = 10$  and  $N = 100$ ), and the magnitude of the effect was low (e.g.  $E_{\text{True}} < 10$ ), fitted parameters overestimated the effects.

Accuracy of the sigmoid model was very high when there was a large number of sampling surveys, due, in large part, to the fact that the entire sigmoid response was evident in the time series. Shorter time series from the *After* period (e.g.  $S_{\text{After}} = 5$ ) and/or few replicates (e.g.  $R = 1$ ) yielded poor estimates of effects (mostly overestimates) because there had not yet been sufficient time to obtain good

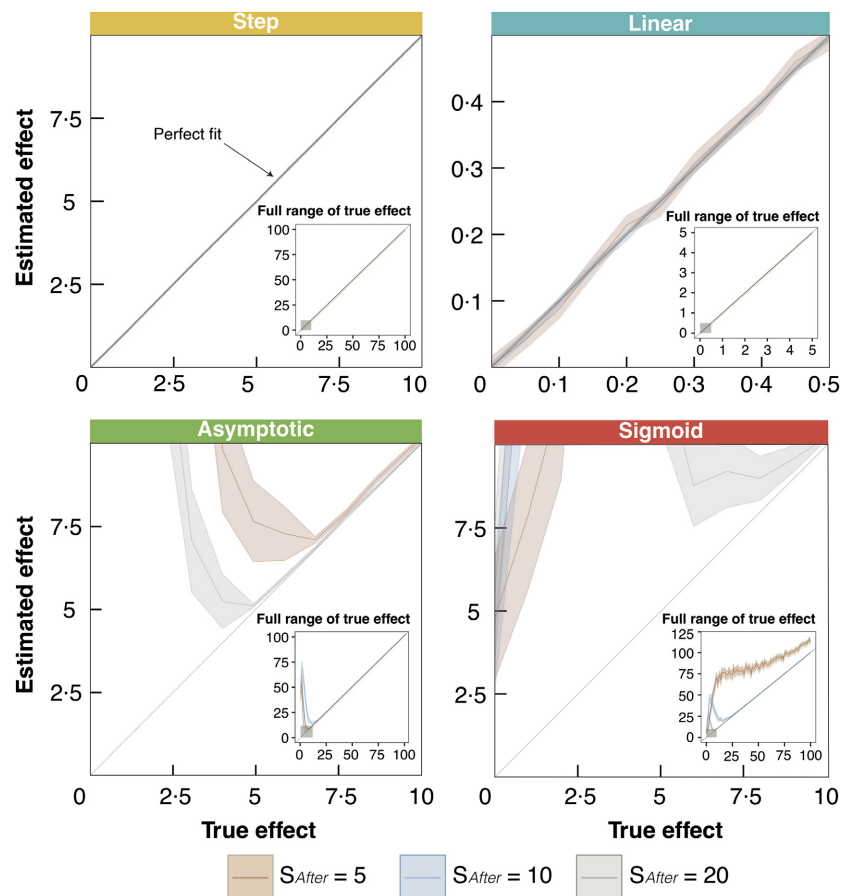
estimates of the long-term sigmoid behaviour (Figs 4 and S4 in Appendix S3).

#### APPLICATION OF PROGRESSIVE-CHANGE BACIPS TO REAL DATA

We successfully applied the Progressive-Change BACIPS to data from three case studies. Following the oil spill, the reproductive success of sea birds declined abruptly (Fig. 5, Case study 1), resulting in selection of the step-change model ( $\omega_{\text{Step}} = 98.2\%$ ; the next best model, linear, had  $\omega_{\text{Linear}} = 1.6\%$ ). The step-change model estimated a decrease of 0.50 fully grown chicks per nest (95% CI: 0.34–0.67;  $P$ -value = 0.006) after the oil spill.

Highway construction, in contrast to the abrupt decline in sea birds in response to the oil spill, caused a gradual decline in the abundance of great bustards. The linear model was best supported by the data ( $\omega_{\text{Linear}} = 64.8\%$ ; the next best model, sigmoid, had  $\omega_{\text{Sigmoid}} = 35.2\%$ ), which produced an estimated rate of decrease of 33.12 individuals per year (95% CI: 21.17–45.06;  $P$ -value = 0.0001) following highway construction (Fig. 5, Case study 2).

The establishment of the FRA (Fig. 5, Case study 3) increased fish density, and the response was best described by a sigmoid model ( $\omega_{\text{Sigmoid}} = 92.9\%$ ; the next best model, linear, had  $\omega_{\text{Linear}} = 7.1\%$ ), indicating that fish density eventually increased by 12.54 fish per  $\text{m}^2$  (95% CI: 10.7–14.4;  $P$ -



**Fig. 4.** Estimated effects (solid lines with  $\pm 95\%$  CI indicated by shading) based on 1000 simulations for different combinations of sampling surveys ( $S_{\text{After}}$ ) and true effects ( $E_{\text{True}}$ ) for four models fit to data generated from the same underlying process: step-change, linear, asymptotic and sigmoid. The black line is the 1:1 line and indicates perfect fit between the estimated effect and the true effect ( $E_{\text{Estim}} = E_{\text{True}}$ ). Consistent deviation of  $E_{\text{Estim}}$  from this expectation represents bias. Effects for the step, asymptotic and sigmoid models, are all based on the change in the difference from Before to the long-term difference After the intervention (i.e.  $M$ ). Effects for the linear model are defined as the estimated (or true) rate of change (i.e.  $r$ ). All results were obtained for  $N = 10$ ,  $R = 10$ ; see Appendix S3 for results obtained by varying  $N$  and  $R$ . For the asymptotic,  $L = 1$ . For the sigmoid model,  $K = 5$  and  $L = 5$ .

value =  $2.69 \times 10^{-16}$ ) and that the time after enforcement required to reach half of this maximum value was 3.49 years (95% CI: 3.18–3.8;  $P$ -value <  $2.69 \times 10^{-16}$ ).

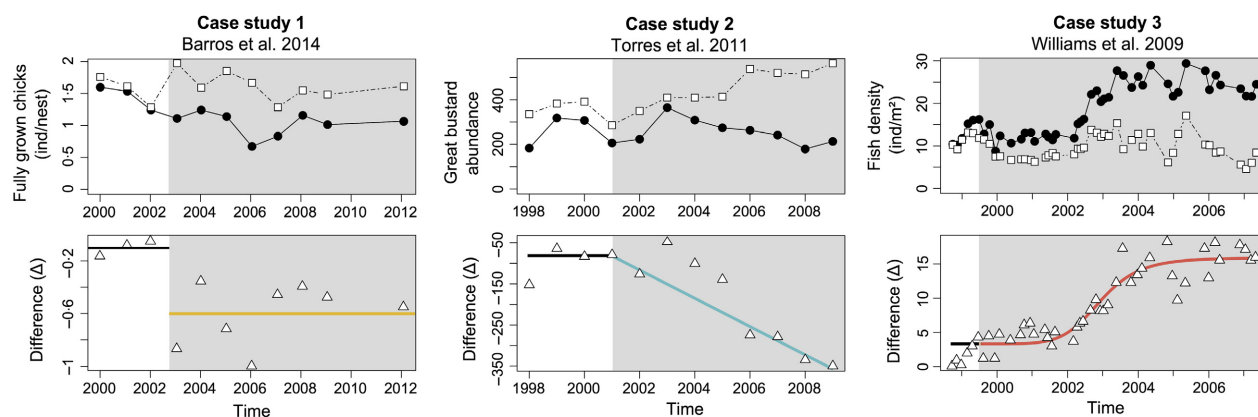
## Discussion

To provide robust inferences for evidence-based management, it is important to develop sampling designs and statistical tools that can provide reliable measures of the effects of interventions on ecosystems. BACIPS designs are particularly powerful tools that allow practitioners to quantify effects of site-specific environmental interventions and distinguish them from other sources of spatiotemporal variation (Stewart-Oaten & Bence 2001). BACIPS designs are generally implemented to analyse step-change responses of the system (as in Case study 1: Barros, Alvarez & Velando 2014). However, we argue that in some cases, *a priori* assumption of an immediate response of the system may be misleading if the ecological consequences arise progressively or are delayed (e.g. Fig. 5; Case studies 2 or 3; Williams *et al.* 2009; Torres *et al.* 2011).

As an improvement, we propose the Progressive-Change BACIPS, which can be used to model a diversity of induced responses, including a step-change. The Progressive-Change BACIPS is therefore more flexible than the original BACIPS formulation, which likely increases the quality (precision and accuracy) of the effect size estimates and the statistical power (i.e. the ability to detect an effect) because the data are used to inform the form of the final selected model, rather than having the form of the model imposed on the data. For example, applying the step-change model (i.e. using the original BACIPS approach) to the data in Case study 2 (Fig. 5) did not lead to detection of a change in great bustard abundance from before to after highway construction ( $E_{\text{Estim}} = -111.43$ ; 95% CI: 119.9 to  $-152.67$ ;  $P$ -value = 0.10), whereas the model selected using the Progressive-Change BACIPS yielded a

significant linear decrease of  $-33.2$  individuals per year (95% CI:  $-21.17$  to  $-45.06$ ;  $P$ -value < 0.001). Similarly, for Case study 3 (Fig. 5), a step-change model, although significant, greatly underestimated the change in fish density ( $E_{\text{Estim}} = 7.98$ ; 95% CI: 1.69–10.58;  $P$ -value = 0.0008) compared to the more likely sigmoid model ( $E_{\text{Estim}} = 12.54$ ; 95% CI: 10.7–14.4;  $P = 2.69 \times 10^{-16}$ ). Applying the Progressive-Change BACIPS can thus help inform better management.

To our knowledge, no time-dependent analytic approaches have yet been used to analyse BACIPS data, although three related approaches warrant discussion. Box & Tiao (1975), one of the most cited and influential environmental intervention analyses, fit a specific model to ozone in Los Angeles using a Before-After design (i.e. without Control data). Their time-series model included both a step-change (due to a highway diversion) and a continuous change (due to a modification to new car engines). Thus, Box & Tiao (1975) implemented a more flexible approach to effects, although the data they used was not based on a BACIPS design and they did not compare competing models. Bence, Stewart-Oaten & Schroeter (1996) applied a predictive BACIPS approach to quantify kelp responses to the effluent of a nuclear power plant. Instead of modelling a time-dependent response, they modelled the effect as a function of the Control site, assuming that effect sizes might vary with the overall quality of the environment: they found larger effects when kelp abundance was greater at the Control site (i.e. during better years). Although more flexible than the classic BACIPS model, Bence *et al.*'s approach still asserts a step-change, but in which the size of the change varies as a function of environmental covariates. Interestingly, Torres *et al.* (2011), which provided the data in our Case study 2, estimated annual population changes of great bustard by comparing annual changes in population size at the Control and Impact sites. Because their approach assumed this rate of change was constant (i.e. that the total effect accumulated in



**Fig. 5.** Application of Progressive-Change BACIPS to three real-world data sets. Case study 1: effect of an oil spill on bird reproductive success (Barros, Alvarez & Velando 2014). Case study 2: effect of a highway construction on great bustard abundance (Torres *et al.* 2011). Case study 3: effect of a fish replenishment area (a type of marine protected area) on fish density (Williams *et al.* 2009). The top panels show the response variable at the Impact (black circles) and Control (white squares) sites Before (white background) and After (grey background) the intervention (Step 1 from Fig. 1). The bottom panels give the time series of differences (white triangles) as well as the fit of the best model based on the Progressive-Change BACIPS method (Steps 2–3 from Fig. 1). The coloured lines in the bottom panels represent the best model to the After differences (step-change model = yellow; linear model = blue; sigmoid model = red), and the solid black lines indicate estimated average differences in the Before period ( $\Delta_{\text{Before}}$ ) based on the best model.

time), their approach is very similar to the Progressive-Change BACIPS presented here. However, they did not compare competing models, and their approach therefore required *a priori* selection of a model which is less informative and may introduce a bias if the wrong model is used (Burnham & Anderson 2002). Of course, our analysis, which found strong support for a linear change, adds support for their approach.

Repeated application of Progressive-Change BACIPS to simulated data under various combinations provided insights on the statistical reliability of the method. The ability of the Progressive-Change BACIPS to correctly quantify the change induced by an intervention will depend on several factors, such as those related to the initial state of the ecosystem (initial density or some other response variable), the type of response induced (pattern of change), the magnitude of the change (effect magnitude), or the sampling effort used to collect data (number of replicate samples within a site and number of sampling dates *After*). However, this is true for any other statistical approach as those factors affect the statistical power of any test.

We show that when effects follow simple dynamics (e.g. step or linear change), the correct model (i.e. model fitted to data generated from the same underlying process) is always selected among the set of candidate models and yields accurate estimates of the effect, even for conditions expected to be challenging (e.g. high initial density, low effect magnitude, few replicates and sampling dates *After*).

Of course, choosing the correct model will be more difficult when spatiotemporal variation is high because it will add additional noise to the underlying relationships that are fitted (e.g. Osenberg *et al.* 1994, 2006). Although we did not include true spatiotemporal variation in the difference in our simulations, we did vary the sampling intensity ( $R$ ) at each survey. Because the estimated difference between the Control and Impact sites will be determined by both the sampling error (a function of  $R$ ) and spatiotemporal variation, the effects of adding spatiotemporal variation are analogous to effects we observed by reducing  $R$  (and indirectly by increasing the initial density, which increases the difficulty in estimating density at a site). Clearly, adding additional variation via spatiotemporal uncertainty will add to the challenges in discerning between competing models. However, this is not specific to the Progressive-Change BACIPS and applies to all existing impact assessment designs.

Inference is even more challenging when dynamics are complex (i.e. with more parameters, such as asymptotic and sigmoid models), and requires: (i) successfully identifying the correct model among a set of candidate models and (ii) using that model to derive reliable estimates of the effects. Successful application of the Progressive-Change BACIPS to complex dynamics required conditions that produce evident patterns in the time series (i.e. high number of replicates and sufficient sampling dates to cover the entire trajectory of change). In fact, sparse data sometimes led to the selection of simple (incorrect) models over more complex (true) models. Step-change and linear models, respectively, outperformed asymptotic and sigmoid models when initial density and effect magnitude were high, and number of replicates and sampling dates *After* the

intervention were low. This lack of support for the true models described above is not surprising and illustrates that the complexity of models should be compatible with the amount of information available. In our case, the use of complex – and likely ill- or over-fitted – models on sparse data led to spurious outcomes. Conversely, simpler models, although not reflecting the true pattern of change, may provide more reliable prediction (Ludwig & Walters 1985). These simple models, when applied to sparse data, may provide coarser, but more reliable, estimates of the effect of an intervention (Anderson *et al.* 2001). Thus, and this is generic to any statistical modelling approach, it is not always desirable to rely on one single *a priori* model, illustrating the benefits of BACIPS analyses that compare several plausible models.

The sigmoid model was sometimes correctly chosen, but still gave biased estimates of effects (Fig. 4): for example, when the *After* sampling was restricted to a few surveys. However, this model was the only model that allowed the differences to accelerate in the early portion of the *After* period (i.e. from  $S_{\text{After}} = 0$  to  $S_{\text{After}} = 5$ ). This acceleration in the times series of the differences led to selection of the sigmoid model, but the short-term series precluded the sigmoid model from estimating the asymptote with great confidence. It is important to note that these conditions correspond to situations in which no model could have provided good estimates of the long-term effects – the data were inadequate. Instead, a model focusing on short-term effects (e.g. an exponential change in the differences) might have been more appropriate. This example emphasizes the critical importance of appropriate selection of competing hypothesis (models) based on the quality of the information available and reasonable initial assumptions when basing inference on multiple models.

Our results show that the performance of the Progressive-Change BACIPS proposed here is affected by the induced response, the magnitude of the change and the sampling effort used to collect data. Because the first two cannot be controlled by investigators, a good understanding of the system is required to develop a suitable sampling strategy (i.e. representative response variable, appropriate Control site) and adjust sampling effort (Stewart-Oaten, Bence & Osenberg 1992; Stewart-Oaten & Bence 2001; Osenberg *et al.* 2006). Sampling effort can be optimized by varying the size (or number) of the sampling unit (e.g. transect, quadrat) and the temporal distribution of the sampling dates *Before* and/or *After*. Of course, these sampling decisions will depend on the pattern of spatial and spatiotemporal variation and the way in which the Control and Impact sites covary in time. Having adequate *Before* data (ideally through regular long-term monitoring of a system for unplanned interventions and specifically designed data collection for planned interventions) to quantify these patterns of variation is essential to implementing a BACIPS study (Stewart-Oaten 1996; Stewart-Oaten & Bence 2001). However, obtaining *Before* data is challenging, especially in the case of unexpected events (e.g. oil spill), and this has greatly limited the application of BACIPS designs (Osenberg *et al.* 2011). Despite this limitation, investigators have implemented BACIPS designs in many contexts: for example, obtaining



Before data from previously established long-term monitoring programs (e.g. Barros, Alvarez & Velando 2014). By taking advantage of these opportunities (for unplanned interventions) or collecting Before data (for planned interventions), we can greatly improve the reliability of inferences about human effects on ecosystems.

Sound environmental impact assessment also requires appropriate statistical models. The Progressive-Change BACIPS presented here is an example of a highly flexible, and easy to implement, approach that allows the detection and quantification of an intervention through time. We hope the Progressive-Change BACIPS will encourage the development of better designed impact assessments, in turn helping practitioners derive more reliable conclusions and improve ecosystem management.

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## Data accessibility

R scripts: uploaded as Supporting Information (Appendix S1).

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## Supporting Information

Additional Supporting Information may be found online in the supporting information tab for this article:

**Appendix S1.** R script to perform Progressive-Change BACIPS analysis.

**Appendix S2.** Ability of the Progressive-Change BACIPS to identify the correct model among the set of candidate models under varying conditions.

**Appendix S3.** Ability of the Progressive-Change BACIPS fitted with the correct model to estimate the true effect under varying conditions.